

Breaking the Trusted Server Assumption in Federated Learning through Symmetric Dual-Sided Scoring

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Abstract—Although Federated Learning (FL) enables privacy-preserving load forecasting in smart grids, current security frameworks suffer from the dogmatic “trusted server” fallacy. While client-side Byzantine defenses are widely studied, the central aggregator remains a critical blind spot and a single point of failure against server-side breaches. To break this dependency, this paper introduces a novel Symmetric Dual-Sided Scoring Framework, pioneering a bidirectional security paradigm for resource-constrained environments. Instead of relying on unidirectional trust, our architecture enforces a mutual, data-blind parameter audit loop deployed symmetrically at both the aggregator and edge layers. Utilizing weights similarity, the framework secures the system from both ends: downstream, low-power smart meters autonomously audit incoming global models to detect and recover from server exploits; upstream, the aggregator tracks cumulative client reputations to filter out distributed edge poisoning. Extensive evaluations on real-world Ausgrid and Meteostat datasets prove that our dual-sided defense successfully mitigates up to an 80% client corruption rate and full root-level server breaches. Maintaining an ultra-low linear computational complexity ($O(n)$), this work demonstrates that robust, mutual auditability can be seamlessly integrated into hardware-restricted IoT grid infrastructures.

Index Terms—Federated learning, mobile edge computing, poisoning attacks, defense robustness, wireless security.

I. INTRODUCTION

Smart grids leverage bidirectional communication and data analytics to optimize and balance real-time energy demand, driven by renewable integration and electric vehicle deployment. However, this vast IoT structure creates critical security and privacy challenges due to continuous consumption monitoring. Federated Learning (FL) mitigates these privacy concerns by enabling decentralized deep learning, ensuring raw smart meter data remains local while only sharing model updates.

Despite protecting data privacy, FL introduces severe vulnerabilities to data and local model poisoning attacks, where malicious actors manipulate updates to distort global regression forecasts, potentially disrupting grid operations or triggering blackouts. While State-of-the-Art countermeasures offer Byzantine-robust aggregation algorithms, they predominantly rely on the restrictive assumption of an inherently “trusted server”, focusing exclusively on filtering static, client-side anomalies. This overlooks advanced persistent threats where the central server itself is transiently compromised via memory exploits or buffer overflows. Such root-level attacks grant an omniscient vantage point to bypass traditional statistical filters. The impact of server-side model poisoning on critical

infrastructures forecasting remains a significant blind spot that this work aims to quantify and resolve.

To address these gaps in the literature and secure FL architectures against advanced adversaries, our contributions in this article are threefold :

- **Pioneering a Symmetrical Dual-Sided Security Paradigm:** We introduce a novel, bidirectional scoring framework that shifts the FL security paradigm from absolute central trust to mutual auditability.
- **Comparative Impact Analysis of Asymmetric Threat Vectors:** We conduct an in-depth comparative study quantifying the structural degradation induced by Data poisoning versus Model poisoning. Crucially, we isolate and model a surgical server-side exploit, empirically demonstrating its severe performance deterioration compared to traditional client-side anomalies.
- **Resource-Constrained Edge-Ready Validation:** We design our dual-scoring mechanism to achieve a strictly linear computational profile ($O(n)$). Through extensive experiments on real-world smart grid datasets, we prove that our framework successfully mitigates up to 80% edge corruption and full server breaches while respecting the tight hardware constraints of low-power IoT devices.

II. STATE OF THE ART (SOTA)

A. Attacks on Federated Learning Model

The vulnerabilities of FL architectures are generally exploited through two main vectors : Data poisoning and Model poisoning.

Data poisoning occurs at the local storage level of edge nodes. Attackers manipulate local consumption data to deceive the model or disrupt its training process. A common approach is the injection of statistical perturbations, such as the Additive Noise Attack (e.g., Gaussian or Uniform noise) [3] or the Scaling Attack [3], which artificially amplifies or attenuates energy consumption values to bias demand forecasting.

Unlike Data manipulation, Model poisoning targets the parameters or gradients directly before they are uploaded to the central aggregator. These attacks tend to be significantly more significant. In a brute-force scenario, such as the Extreme Value Deviation Attack (EVDA) [4], local weights are replaced with arbitrary extreme constants to break convergence. Alternatively, noise can be directly injected into the local gradients.

B. Existing defense mechanisms

To mitigate these distributed threats, the research community has established countermeasures that operate either on the client side during training or at the central server during aggregation.

The core defense of FL architectures traditionally resides at the central aggregator to filter out Byzantine clients.

1) *Statistical and Geometrical Estimators*: Classic robust aggregation operators replace the standard federated averaging (FedAvg). Algorithms such as Trimmed Mean (T-Mean) and Bulyan [6], which couples Multi-Krum and T-mean, strip away extreme parameter values before executing aggregation.

2) *Clustering Frameworks*: Methods like Centroid-Based Anomaly-Aware FedAvg (CBAA-FedAvg) [2] group local updates based on spatial similarity (e.g., cosine similarity). They dynamically penalize highly correlated updates, making them effective against coordinated Sybil attacks.

3) *Heuristic and Scoring Models*: Advanced systems track historical data or distributions shifts using Cumulating KL-Divergence [6]. Furthermore, frameworks like Zeno [7] compute a validation score for each client based on its actual performance step on a dedicated server-side validation dataset, selecting only the top- k updates.

Despite the abundance of literature on FL security, existing models suffer from two foundational limitations that render impractical for real-world deployments :

- **The "Trusted Server" Fallacy**: The majority of current state-of-the-art defenses (including Krum, Bulyan, RFA and FoolsGold) implicitly assume a fully secure, uncompromised central server. They focus exclusively on mitigating client-side Byzantine disruptions. However, the central aggregator represents a critical single point of failure.
- **Heavy Server Dependencies and Client Resource Blindness**: Most scoring of defensive validation systems (e.g., Zeno, Weighted Aggregation) require the central server to hold a clean, global verification dataset. This requirement violates strict user privacy constraints and data siloing. Conversely, complex client-side defenses like SHAP demand extensive computational overhead, making them entirely incompatible with the strict hardware limitations of low-power IoT edge nodes and smart meters.

To the best of our knowledge, little to no research explicitly investigates the significant impact of server-side model poisoning, nor does the literature offer a lightweight, decentralized scoring-based defense mechanism deployed directly on the nodes to protect the edge layer when the central authority itself cannot be trusted. This work bridges this crucial gap.

III. SYSTEM MODEL

To formalize the context of the study, we define a decentralized ecosystem, encompassing the network topology, the client participation workflow and the specific neural network architecture.

A. Network Architecture and Client Sampling

The ecosystem consists of a central aggregator (server) and a pool of N decentralized edge nodes, represented by IoT-enabled devices and residential smart meters. The primary objective is to collaboratively train a global machine learning model without centralizing private consumption data. The training process is structured into sequential federated rounds $t \in \{1, 2, \dots, 20\}$.

At the beginning of each round t , the central server does not communicate with the entire network to preserve bandwidth and adapt to device availability ; instead, it performs a client sampling step. A subset of clients K is randomly selected. The server broadcasts the current global model weights W^t to these selected smart meters. Each sampled node then executes local training for 15 epochs using its local energy consumption dataset, generating an updated set of local weights w_k^{t+1} .

B. Machine Learning Model Architecture

Designed to balance predictive accuracy with the strict hardware limitations of low-power IoT devices, the network architecture is structured as a sequential deep learning model for regression. The multi-dimensional time-series consumption is initially processed by a Flatten layer to map the input tensor into a one-dimension vector, which is then fed into two consecutive fully connected layers of 32 neurons each utilizing ReLU activation functions to extract non-linear load features. To prevent overfitting, and enhance generalization across heterogeneous smart meters, a Dropout layer with a 20% rate ($p = 0.2$) is integrated immediately after these dense layers. Finally, the features are mapped to a three-dimensional output vector ($\mathbb{R}^3$) via a Fully Dense layer equipped with a Linear activation function, allowing the model to perform continuous multi-horizon energy forecasting on grid state regression tasks before transmitting the local weight updates back to the aggregator.

IV. THREAT MODEL

To systematically evaluate the security of the proposed federated learning framework, we formalize our threat model into four key dimensions : the adversary's goal, knowledge, capability and specific operational strategies.

A. Adversary's goal

The primary objective of the adversary in this smart-grid use case is to disrupt the regression performance of the model (e.g., load forecasting or state estimation). Specifically, the attacker aims to degrade overall accuracy by driving up the regression error metrics (such as Mean Square Error) to induce poor grid optimization, forcing incorrect energy distribution or false alerts.

B. Attacker's knowledge

We assume the adversary operates under a realistic gray-box knowledge model. At the edge layer, the attacker possesses full visibility over the compromised smart meters, including their local consumption data and the architecture of the local FL

model. However, they lack direct knowledge of other clients' private datasets or the exact mathematical aggregation weights computed by the server.

At the server layer, in the event of an intrusion, the attacker has enough system-level insight to identify memory structures containing the global model parameters, enabling targeted exploitation (e.g., locating specific offsets of the final layers in memory).

C. Attacker's capability

The adversary's operational capabilities differ heavily between the resource-constrained edge layer and the heavily guarded central server :

- **Client-Side Capabilities:** Because smart meters and IoT edge nodes feature minimal security due to hardware and cost constraints, they are easily exploitable. The attacker can execute attacks continuously across all federated rounds without being easily detected by the central infrastructure.
- **Server-Side Capabilities:** The central aggregator benefits from advanced monitoring and immediate intrusion detection. Consequently, the attacker's capability on the server is highly constrained in time. They cannot maintain a long-term presence and are rapidly ejected from the system, limiting their window of execution to either a single round or a few consecutive rounds before mitigation.

D. Attacker's strategy

Based on the aforementioned dimensions, the adversary deploys three distinct attack strategies to benchmark the framework's resilience :

1) *Strategy 1: Continuous Distributed Edge Poisoning:* The attacker persistently injects corrupted data or local weights throughout the training. It benchmarks three severity levels: Low (5% – 15%) for subtle anomalies, Medium (20% – 55%) for coordinated breaches, and High (60% – 80%) for massive zero-day exploits.

2) *Strategy 2: Surgical Server-Side Partial Model Poisoning:* Exploiting a transient server exploit, the attacker surgically overwrites only the final layers. This stealthy intervention is short-lived, lasting only a few consecutive rounds before detection.

The formalization of our threat model underscores the critical necessity for a hybrid, bidirectional security architecture capable of addressing the structural blind spots inherent to traditional centralized defenses.

First, to counter the risk of transient server-side compromise, it is imperative to decouple edge security from absolute central trust. This establishes a clear requirement for a lightweight node-side self-defense mechanism. Second, the system must address the challenge of continuous, low-footprint client-side poisoning. Consequently, the framework requires a dynamic reputation scoring table maintained at the server level. Finally, the proposed framework has to fulfill system resilience needs and adapt to limited hardware computational resources in critical infrastructures.

V. EDGE DEFENSE PROPOSITION : DUAL-SIDED ENTITY SCORING

A. Intuitive idea

The core intuition behind the proposed mechanism relies on structural and behavioral continuity across consecutive federated learning rounds. In a typical FL environment deployment, global model parameters evolve smoothly as convergence approaches. A malicious intervention inevitably introduces sharp discontinuities or performance collapses.

Our framework mitigates these threats via a bidirectional auditing pipeline :

- **Server-to-Node Auditing:** Each smart meter independently evaluates the relationship between the local-model sent to the central server at round $t - 1$ and the new global model received at round t . If a server compromise is detected, the node discards the update.
- **Node-to-Server Auditing:** Simultaneously, the central aggregator maintains a reputation profile for each edge device. By monitoring incoming local weight updates over time, the server calculates a client reliability score, preventing poisoned edge parameters from polluting the global model aggregation.

B. Scoring methods

Because the scoring architecture is symmetric, both entities compute the trustworthiness score $S \in [0; 1]$. We can utilize four distinct indicators to form the baseline of these scores :

- **Euclidean Distance:** Measures the raw geometric divergence between parameter vectors.
- **Jensen-Shannon (JS) Divergence:** Quantifies the statistical distance between weight distribution profiles.
- **Empirical local validation (Node-Only):** Measures the Mean Squared Error (MSE) of the incoming model on a micro, hidden local dataset.
- **Weights Similarity:** Evaluates cosine similarity, magnitude similarity and sign similarity to detect adversarial attacks or coordinated Sybil vectors :

$$SIM_{cosine} = \frac{W_t \cdot W_{t-1}}{|W_t| |W_{t-1}|} \quad (1)$$

$$SIM_{sign} = \frac{1}{Card(W_t)} \cdot \sum_{i \in W_t} 1_{sign(W_t^{(i)}) = sign(W_{t-1}^{(i)})} \quad (2)$$

$$SIM_{magnitude} = Pearson(W_t, W_{t-1}) \quad (3)$$

$$S = \frac{1}{3} \cdot \left(\frac{SIM_{cosine} + SIM_{magnitude} + 2}{2} + SIM_{sign} \right) \quad (4)$$

C. Decision rule

Node-Side Rule (Server Filtering & Recovery): Upon receiving the new global model W_t , the node computes $S_{node} = \text{Score}(W_t, W_{t-1})$. A static threshold $\tau \in [0; 1]$ is applied :

- If $S_{node} \geq \tau$, the global model is accepted for local training.
- If $S_{node} < \tau$, the node flags a server-side breach, discards W_t and triggers a state recovery routine. The smart meter rolls back to W_{t-1} and continues training in a sandboxed local state, protecting its local architecture from corruption.

Server-Side Rule (Client Filtering & Robust Aggregation): Upon receiving a local-model update w_k^t from a sampled client k , the server computes its reliability score $S_{server} = \text{Score}(w_k^t, w_k^{t-1})$. The aggregator applies a filtering threshold $\tau \in [0; 1]$:

- If $S_{server} < \tau$, the client update is immediately classified as an outlier (poisoned by data or local-model tampering) and excluded from the aggregation pipeline.
- For all compliant clients ($S_{server} \geq \tau$), the server dynamically adjusts their aggregation weights proportional to their reliability, mitigating the influence of borderline updates before producing W_{t+1} .

D. Computational Complexity

Because the framework excludes heavy validation data inference or auxiliary models (such as GANs or VAEs), the complexity of calculating similarities (SIM_{cosine} , SIM_{sign} , and $SIM_{magnitude}$) scales purely linearly with the number of model parameters, yielding a time complexity of $O(n)$, where n is the total number of weights. Given our optimized grid model architecture (consisting of two dense layers of 32 neurons), n is highly compact. Consequently, the scoring pipeline introduces negligible processing latency on smart meters and allows the central server to rapidly audit hundreds of concurrent client streams in parallel without degrading real-time grid operations.

Moreover, the proposed security layer introduces zero communication overhead. Unlike alternative defenses that require potential auxiliary data exchanges, multi-party cryptographic protocols, or additional verification rounds, our scoring functions operate entirely on the parameters already transmitted during standard FL operations. By piggybacking exclusively on the existing upstream and downstream communication flows, the framework guarantees robust bidirectional security without consuming extra network bandwidth or increasing grid telemetry latency.

VI. EXPERIMENTATION SETUP AND MAIN RESULTS

To demonstrate the efficiency of our symmetric dual-sided scoring defense, we conduct extensive experiments using real-world smart grid and meteorological data. This section details the dataset characteristics, feature engineering and preprocessing pipeline.

A. Experimentation setup

1) *Dataset Description:* We utilize the Ausgrid dataset [9], which provides half-hourly smart meter measurements from 300 residential households located in New South Wales, Australia, spanning from July 2010 to June 2013. The dataset captures two primary electrical components : general electricity consumption and controlled load. In this study, these two components are aggregated into a single total demand metric to reflect the actual, holistic electricity requirement of each household.

To expand the predictive capability of our models, we integrate localized weather data sourced from the Meteostat platform [10] for the identical geographical region and timeframe. Because weather conditions strongly influence photovoltaic (PV) generation efficiency and indirectly dictate residential heating/cooling consumption, meteorological parameters serve as high-value predictors for our uniform 30-minute temporal resolution, the datasets are chronologically aligned and merged based on exact timestamps.

2) *Feature Engineering and Preprocessing Pipeline:* To transform the raw integrated data into a format suitable for our compact deep learning regression model, a multi-step preprocessing pipeline is implemented.

First, raw timestamps are engineered to capture human-activity cycles and seasonal variations. The date component is mapped to a binary weekend indicator, where weekend days are encoded as 0 and weekdays as 1. Furthermore, to prevent the model from interpreting time-of-day as a linear continuum and to accurately preserve its cyclical nature (where 23:30 is adjacent to 00:00), the time component is transformed into two continuous features using trigonometric encoding :

$$Time_{sin} = \sin\left(\frac{2\pi \cdot t}{T}\right), Time_{cos} = \cos\left(\frac{2\pi \cdot t}{T}\right) \quad (5)$$

where t represents the current half-hour interval of the day and $T = 48$.

Second, we compute the presumption as the net energy demand (E_{net}), mathematically formulated as the total household load minus the gross PV generation.

Then, to ensure stable neural network convergence and to satisfy the Independent and Identically Distributed (IID) assumptions required across the federated client nodes, data scaling is performed.

Finally, the processed historical data of each household is chronologically partitioned into three subsets to prevent data leakage during evaluation. The dataset is split into 70% for training, 20% for validation and 10% for testing.

3) *Simulation Setup and Hyperparameters:* To evaluate the resilience of the proposed dual-sided scoring framework, we construct a controlled Federated Learning simulation environment. The network consists of a total pool of N decentralized clients (smart meters). The collaborative training is conducted over a total of $T = 20$ federated communication rounds.

At each round t , a random client sampling rate of 50% of smart meters are dynamically selected to participate in the current training phase. Once selected, each node executes local

optimization for $E = 15$ local epochs before transmitting its parameters updates. The local neural networks are trained using the Adam optimizer with default learning rate configurations. To establish a standard performance baseline in an uncompromised environment, FedAvg (Federated Averaging) is used as the default aggregator operator.

To evaluate the impact of the distributed edge threats described in our Threat Model, we vary the adversarial footprint within the network. We simulate two distinct corruption ratios by designating either 5% (1 malicious client) or 20% (4 malicious clients) of the population as attackers executing continuous data or local-model poisoning.

4) *Baseline Defense Mechanism for Comparison:* To demonstrate the superiority and lightweight nature of our scoring methodology, we benchmark our framework against a comprehensive suite of state-of-the-art server-side and Byzantine-robust aggregation defenses. The selected baselines, with diverse geometric, statistical and clustering paradigms, are shown in table I.

TABLE I
LIST OF BASELINE DEFENSES USED TO EVALUATE OUR PROPOSED DEFENSE MECHANISM

Category	Defenses
Geometric & Distance-Based	Krum, M-Krum [6]
Statistical Estimators	T-Mean [6]
Clustering & Trust frameworks	CLRA [8]

By evaluating our symmetric mechanism against this wide array of established countermeasures under identical threat ratios and server exploits, we provide a holistic validation of its security-to-overhead trade-off critical smart grid operations.

B. Experimentation results

In this section, we present a comprehensive evaluation of our framework. We first analyze the structural impact of the various poisoning strategies against the baseline model. We then evaluate the effectiveness of our proposed symmetric dual-sided scoring defense compared to state-of-the-art countermeasures.

C. Impact of Adversarial Attacks on the Unprotected System

We first benchmark the degradation of the smart grid regression model under the different attack scenarios defined in our Threat Model, using standard FedAvg without any defense. Table II reports the Final Training Loss alongside the testing performance metrics : Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared Error (RMSE).

As observed in table II, continuous client-side model poisoning induces a more severe degradation of regression metrics compared to data poisoning. At a 20% corruption ratio, local-model poisoning drives the test MSE up to $+2,6 \cdot 10^7\%$, whereas data poisoning exhibits a slower error growth due to the inner capacity of FedAvg to smooth out statistical data anomalies over multiple local epochs.

The experimental results clearly validate our threat model assumptions regarding server-side vulnerabilities. Although the server-side total model poisoning only lasts for a single round before attacker ejection, it inflicts an instantaneous collapse of the network profile, resulting in the highest test RMSE (53,36). This demonstrates that a single-round full-model compromise is structurally more damaging to the smart grid's forecasting capabilities than a continuous low-footprint (5%) client-side attack spanning all 20 rounds.

D. Comparative Evaluation : State-of-the-Art Defenses vs. Proposed Scoring Method

To demonstrate the superiority of our symmetric dual-sided Scoring mechanism, we evaluate it against the baseline defenses under three representative threat conditions : a 20% distributed edge compromise, a 80% distributed edge compromise to test the limits of the proposed method, and a surgical server-side attack. Performance is shown in table III and is evaluated using the test MAE.

Under a heavy 20% client compromise, Krum, Trimmed Mean and CLRA manage to reduce the test MAE compared to standard FedAvg. However, our proposed server-side component outperforms them by 86%. This is because our metric tracking functions evaluate the continuity of updates between the last sent model and the received one, successfully catching attackers that trick static geometric distance.

The most critical finding highlighted in table III is the failure of classical State-of-the-Art defenses against the server-side exploit. Because these algorithms assume the server is a fully trusted entity, they are automatically bypassed when the aggregator itself is corrupted. Conversely, our Node-Side scoring rule allows smart meters to autonomously detect the anomalous update, drop the corrupted global parameters, and trigger the state recovery routine. As a result, our framework maintains a stable test MAE of 0,7, close to the uncompromised baseline.

VII. DISCUSSION

While the experimental results validate the efficiency of our symmetric dual-sided scoring framework, several technical dimensions and open challenges warrant further analysis.

A key characteristic of our framework is its reliance on the static trust threshold (τ). Crucially, this boundary is not universally fixed; it is highly sensitive to the underlying operational environment; in a non-IID smart grid scenario, the natural geometric distance between honest model updates widens. Consequently, setting an inappropriate static threshold risks increasing the False Positive Rate, where legitimate updates from atypical consumers are mistakenly discarded. The defense perimeter must therefore be carefully calibrated based on the targeted dataset profile and load variance.

Another critical insight concerns the evolutionary nature of the training process. As the regression model converges over the federated rounds, the structural updates ($W_t - W_{t-1}$) naturally become smaller and more stable. Because our current implementation utilizes a static threshold, this convergence

TABLE II
PERFORMANCE METRICS UNDER VARIOUS ATTACK SCENARIOS AND WITHOUT ANY DEFENSE (FEDAVG AGGREGATION)

Scenario	Training phase MSE Loss	MAE	Test phase MSE	RMSE
Baseline	0,009	0,07	0,01	0,1
Client Data poisoning (5%)	0,009 (+0%)	0,07 (+0%)	0,01 (+0%)	0,1 (+0%)
Client Data poisoning (20%)	0,009 (+0%)	0,06 (-14%)	0,01 (+0%)	0,09 (-10%)
Client Model poisoning (5%)	0,01 (+11%)	2,36 (+3 271%)	35,4 (+353 900%)	5,95 (+5 850%)
Client Model poisoning (20%)	0,3 (+3 233%)	34,91 (+49 914%)	2632,14 (+26 321 300%)	51,3 (+51 200%)
Server Partial Model poisoning	2,15 (+23 788%)	0,13 (+85%)	0,07 (+600%)	0,26 (+160%)
Server Total Model poisoning	151,25 (+1 680 455%)	20,27 (+28 857%)	2847,04 (+28 470 300%)	53,36 (+53 260%)

TABLE III
DEFENSIVE MITIGATION COMPARISON UNDER 20% AND 80% CLIENT POISONING AND SERVER EXPLOIT

Defense	Test Mean Absolute Error (MAE)			
	20% Clients compromise	80% Clients compromise	Global model partial poisoning	Global model total poisoning
Baseline (FedAvg)	0,23	3,82	0,08	18,27
Krum	0,1 (-56%)	14,51 (+279%)	0,08 (+0%)	18,28 (+0%)
Multi-Krum	0,08 (-65%)	2,82 (-26%)	0,08 (+0%)	18,27 (+0%)
Trimmed-Mean	0,09 (-60%)	2,96 (-22%)	0,12 (+50%)	18,36 (+0,5%)
CLRA	0,09 (-60%)	0,33 (-91%)	0,36 (+400%)	18,3 (+0,2%)
Proposed method	0,03 (-86%)	0,21 (-94%)	0,07 (-12%)	0,7 (-96%)

trajectory introduces a late-stage security vulnerability. To mitigate the risk of an attacker exploiting the narrowing variance, future work will explore a dynamic thresholding mechanism that systematically tightens the defense perimeter as the model matures.

Finally, our analysis suggests that the use of only one metric to compute the score might not be optimal against all threat variations. Selective metric coupling could significantly enhance detection precision. For instance, while Euclidean distance excels at detecting brute-force weight scaling, it remains blind to direction-aligned sign flips, which are better captured by similarity metrics.

VIII. CONCLUSION

This paper addressed a critical security gap in Federated Learning environments, explored in a smart-grid use case: the vulnerability of traditional architectures to server-side model poisoning. To resolve this without exceeding the hardware limitations of IoT edge devices, we proposed a Symmetric Dual-Sided Scoring Framework. By deploying an identical, data-blind parameter validation suite at both the central aggregator and individual smart meters, our framework establishes a mutual, localized auditability protocol. Experimental results demonstrate that while State-of-the-Art defenses collapse under server-side exploits, our solution successfully isolates corrupted parameters, filters distributed edge-level poisoning up to an 80% corruption rate, and preserves forecasting performance and stability.

Future work will investigate dynamic thresholding models to neutralize late-stage stealthy exploits during convergence. Additionally, we aim to design a resilient, degraded training mode. This protocol will allow smart meters to autonomously bypass the central aggregator during persistent attacks or

server outages, pivoting to a localized, peer-to-peer training scheme to guarantee continuous, offline learning survival for critical power grid operations.

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